

PROJECT

12

CREATE S3 BUCKETS WITH TERRAFORM

DEVOPS TOOL

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Objective

This project uses Terraform to automate the provisioning of AWS S3 buckets. It demonstrates how Infrastructure as Code (IaC) simplifies resource creation, management, and modification with reusable, version-controlled configurations.

It sets up an S3 bucket and uploads an object using Terraform scripts, showcasing how developers can manage cloud storage resources in a fast, consistent, and scalable manner.

Services

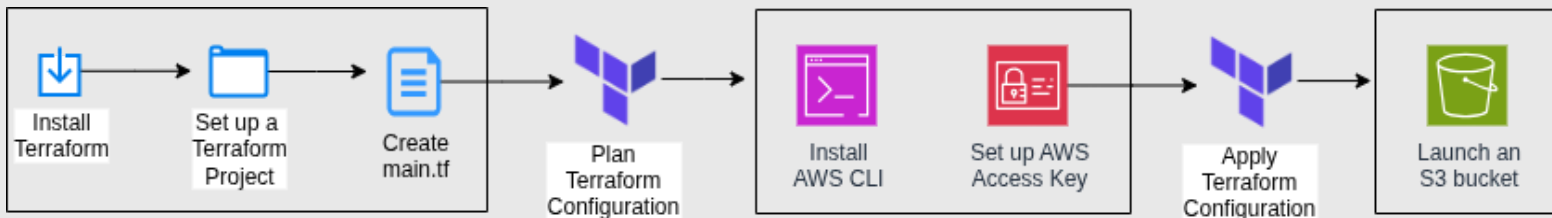
Amazon S3 - Provides scalable object storage for files, media, and application data.

Terraform - Automates infrastructure provisioning using declarative configuration files.

AWS IAM - Manages credentials and permissions to allow Terraform access to AWS.

AWS CLI - Used for configuring credentials and verifying S3 resources.

Architecture Diagram



To begin, install Terraform and set up your project by writing configuration files. Once ready, plan the configuration, install the AWS CLI, and securely set up your AWS access key. Finally, apply the configuration using Terraform to seamlessly launch an S3 bucket in your AWS environment.

Steps to Implement

1. Set up your Terraform project folder:

Create a new directory to hold all Terraform files. Inside it, create a main.tf file where you'll define your infrastructure. Ensure Terraform is installed and ready to use in your system.

2. Define your Terraform configuration:

In main.tf, declare the AWS provider and region for deployment. Then define an aws_s3_bucket resource with a unique bucket name and optional tags. This tells Terraform what to create.

3. Configure AWS credentials for Terraform:

Use aws configure or environment variables to set up your credentials. These allow Terraform to interact with your AWS account. Make sure the IAM user has permissions for S3.

4. Initialize your Terraform project:

Run terraform init inside your project folder to initialize it. This downloads necessary providers and prepares your working directory. You only need to do this once per project setup.

5. Review and apply your configuration:

Run terraform plan to preview what will be created. Then use terraform apply and confirm the changes to create the S3 bucket. Verify the bucket is successfully created in the AWS console.

6. Upload an object to S3 using Terraform:

Add an aws_s3_object resource in main.tf with file path, bucket name, and source. Run terraform apply again to upload the file into the S3 bucket. This automates object storage using code.

Story Time

Terraform (The Cloud Architect's Magic Wand):

Terraform is your magic wand that draws infrastructure from thin air. Just write the spell (config), and it builds the cloud exactly as you wish.

main.tf (The Blueprint Scroll):

The main.tf file is your scroll of instructions for the cloud architect. Every line in it tells Terraform what to summon into existence.

AWS Credentials (The Keys to the Cloud Kingdom):

AWS credentials are your golden keys to enter and build in the cloud realm. Without them, Terraform can't do any of its magic tricks.

terraform init (The Awakening Ritual):

This command wakes up your Terraform project and connects it to the cloud. It gathers the right powers (providers) to cast the spells.

terraform apply (The Summoning Spell):

With a simple apply spell, your bucket appears in the AWS sky. It's like whispering your wishes and watching them come true.

S3 Bucket (The Treasure Vault of the Cloud):

The S3 bucket is your enchanted vault where you can store files, gold, or scrolls. It's secure, always available, and magically scalable.

BY
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THANK YOU!

